Engineer to Excel

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Saveetha Institute of Medical And Technical Sciences

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CO1 AT-3 Technical presentation

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CSA 0504

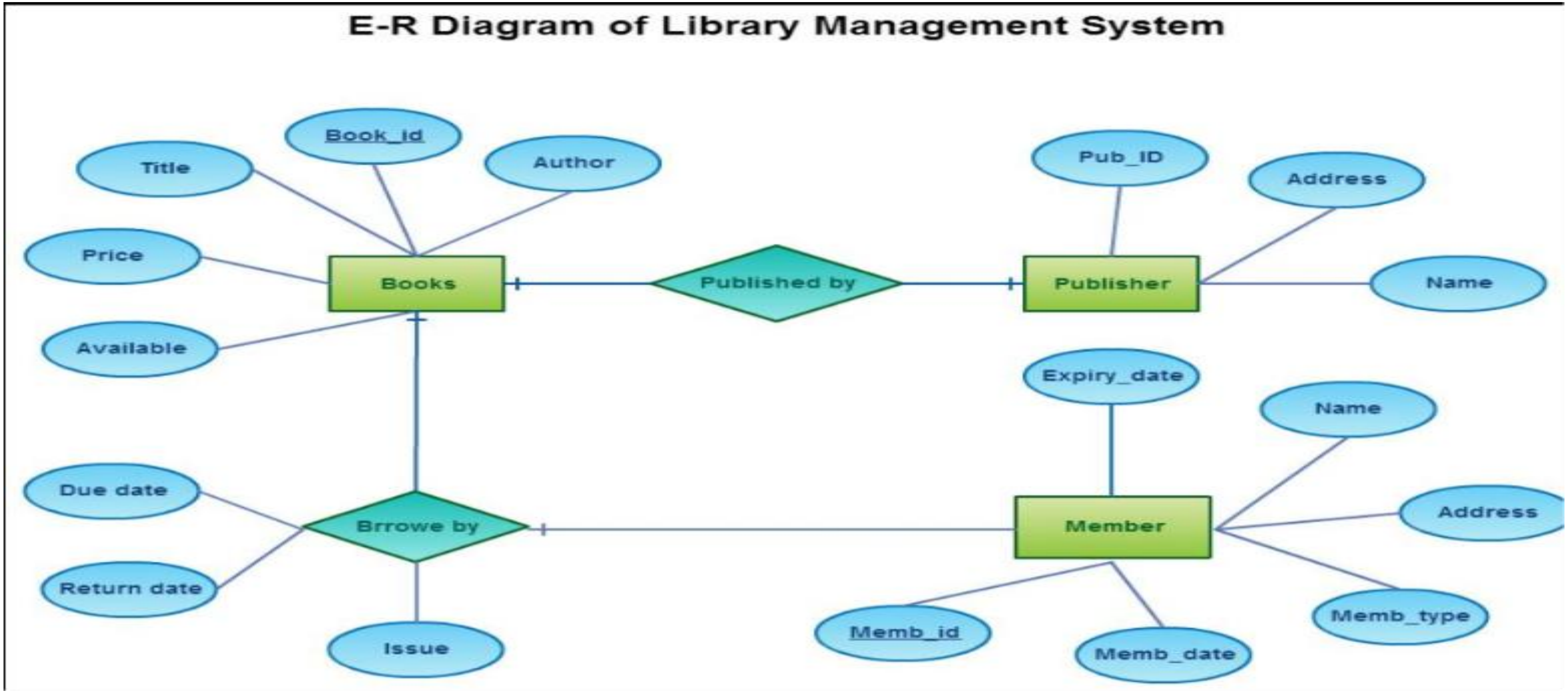
Data base management systems

ER diagram for an online library system

Introduction

- An **Online Library System** is a digital application used to manage library activities efficiently.
- It enables users to **search, borrow, reserve, and return books** online.
- The system maintains records of **books, members, librarians, transactions, reservations, and fines**.
- An **Entity-Relationship (ER) Diagram** is a graphical representation of the database structure.
- It identifies the **entities**, their **attributes**, and the **relationships** between them.
- The ER diagram helps in designing a well-organized and efficient database.

ER Diagaram


E-R Diagram of Library Management System
Book_id
Title
Author
Price
Books
Available
Published by
Publisher
Pub_ID
Address
Name
Expiry_date
Name
Due date
Brrowe by
Member
Address
Return date
Issue
Memb_id
Memb_date
Memb_type

Conclusion

- The ER diagram provides a clear and organized view of the Online Library System database.
- It shows the relationships between books, members, librarians, borrowing, reservations, and fines.
- It helps in designing an efficient and structured database.
- It minimizes data redundancy and maintains data consistency.
- It improves the accuracy of storing and retrieving library information.
- It supports smooth management of book borrowing and return processes.

EER model demonstrating generalization and specialization with a university

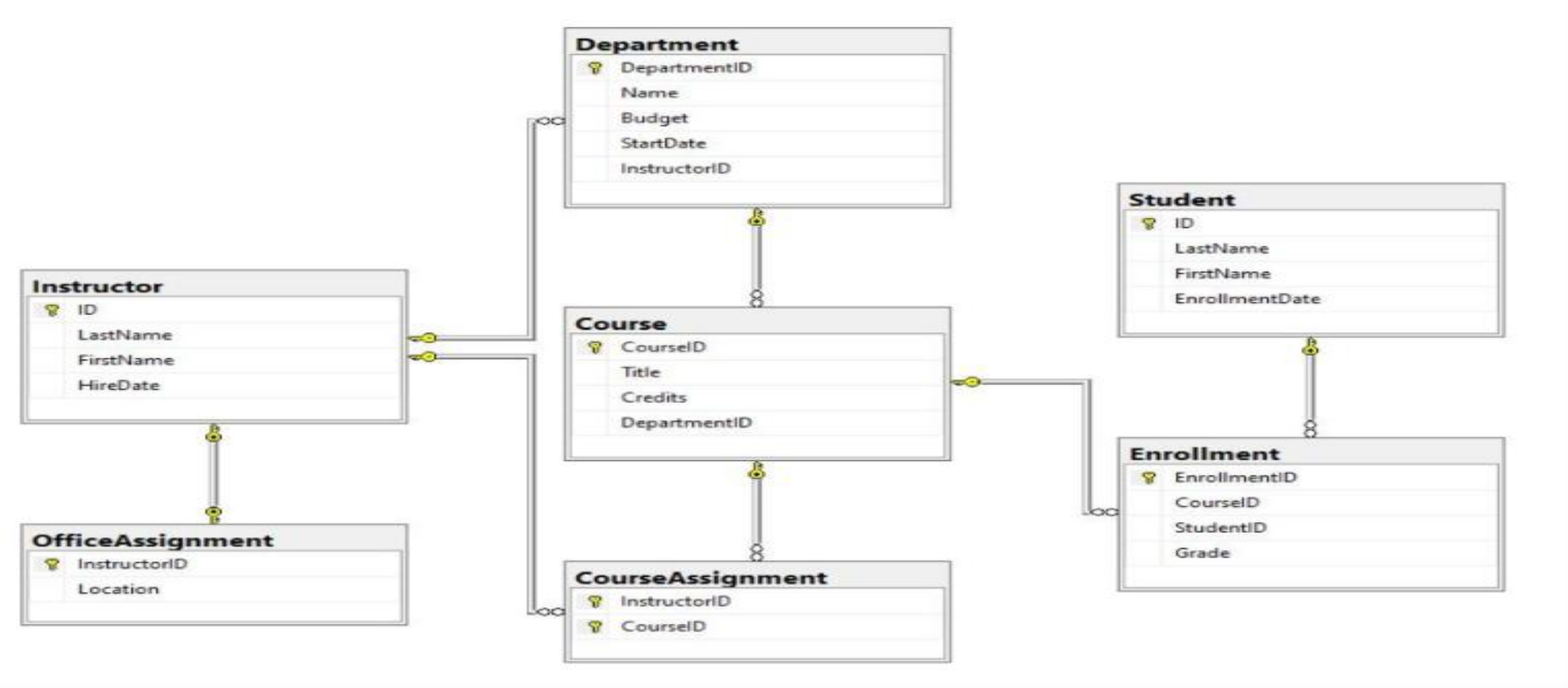
introduction

- The **Enhanced Entity-Relationship (EER) Model** is an extension of the ER model that represents more complex data relationships.

- It introduces advanced concepts such as **generalization, specialization, inheritance, and categorization**. **Generalization** combines similar lower-level entities into a higher-level entity based on their common attributes.

- **Specialization** divides a higher-level entity into lowerlevel entities based on their unique characteristics.

EER diagram for university management



conclusion

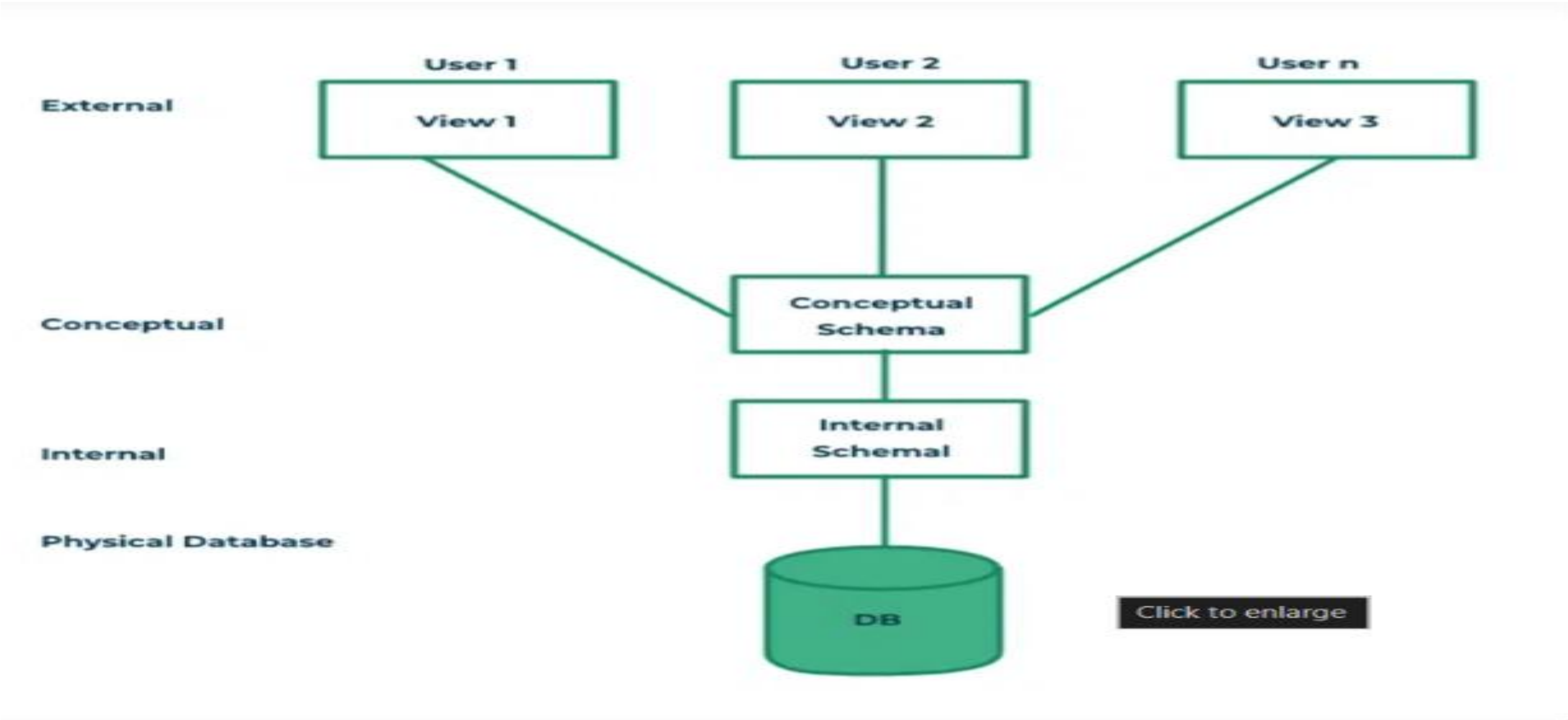
- The **EER model** provides a more advanced and flexible database design than the basic ER model.
- It uses **generalization** and **specialization** to represent real-world university entities effectively.
- It allows common attributes to be inherited by specialized entities, reducing data redundancy.
- It clearly distinguishes between **Students, Faculty, and Staff** while maintaining their relationship with the **Person** entity.
- The model improves data organization, consistency, and accuracy. It simplifies database maintenance and future modifications.
- It supports efficient storage and retrieval of university information.

Data independentand three –schema architecture

Introduction

- **Data Independence** is the ability to modify the database structure without affecting application programs.
- It ensures that changes made at one level of the database do not impact other levels.
- There are **two types of data independence**: **Physical Data Independence** and **Logical Data Independence**.
- **Physical Data Independence** allows changes in physical storage without affecting the logical structure of the database.
- **Logical Data Independence** allows changes in the logical structure without affecting user applications or external views.
- The **Three-Schema Architecture** is a database design framework that separates the database into three levels.

Data independence in dbms –three schema architecture



Conclusion

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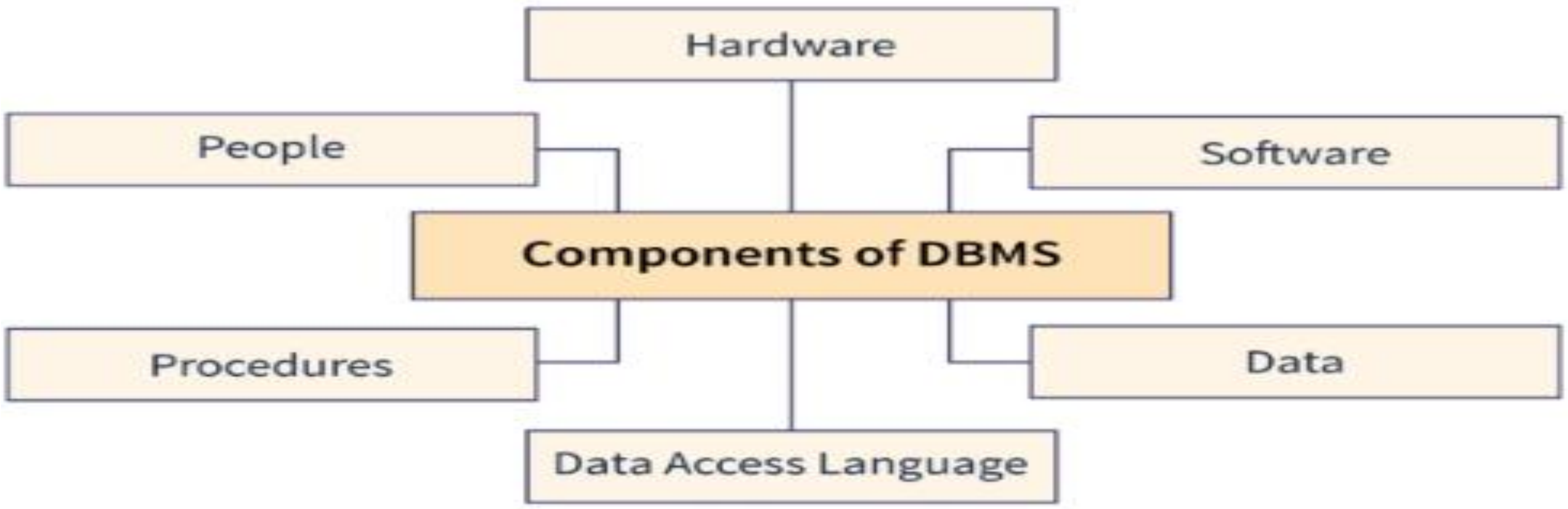
Software components of a dbms

introduction

- A **Database Management System (DBMS)** is software used to create, store, retrieve, and manage data efficiently.
- A DBMS consists of several **software components** that work together to manage database operations.
- These components help users interact with the database securely and efficiently. The **Query Processor** interprets and executes SQL queries submitted by users.
- The **Storage Manager** manages the storage, retrieval, and updating of data in the database.
- The **Database Engine** processes requests and ensures smooth execution of database operations.
- The **Transaction Manager** maintains data consistency and handles concurrent transactions. The **Recovery Manager** restores the database after system failures and ensures data reliability.

Software components of dbms



conclusion

- The software components of a **DBMS** work together to manage database operations efficiently.
- Each component performs a specific function, such as query processing, storage management, transaction control, and recovery.
- These components ensure accurate storage, retrieval, and updating of data. The **Transaction Manager** maintains data consistency and supports concurrent access.
- The **Recovery Manager** protects data and restores the database after failures. The **Security Manager** ensures that only authorized users can access the database.
- The combined operation of these components improves database performance and reliability.

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Comparison of strong and weak entities

Introduction

- **Strong and Weak Entities** are important concepts in the Entity-Relationship (ER) model used in database design.
- A **Strong Entity** is an entity that has its own primary key and can exist independently. A **Weak Entity** is an entity that does not have a primary key of its own and depends on a strong entity for its identification.
- Strong entities are represented by a **single rectangle** in an ER diagram.
- Weak entities are represented by a **double rectangle** in an ER diagram.
- A weak entity is identified using a **partial key** along with the primary key of its related strong entity.
- The relationship between a strong entity and a weak entity is called an **identifying relationship**, represented by a **double diamond**.
- Strong entities can exist independently, whereas weak entities cannot exist without their corresponding strong entity.

Comparison of strong and weak entities in ER modeling

Strong Entity Set	Weak Entity Set
Strong entity set always has a primary key.	It does not have enough attributes to build a primary key.
It is represented by a rectangle symbol.	It is represented by a double rectangle symbol.
It contains a Primary key represented by the underline symbol.	It contains a Partial Key which is represented by a dashed underline symbol.
The member of a strong entity set is called as dominant entity set.	The member of a weak entity set called as a subordinate entity set.

conclusion

- **Strong entities** have their own primary key and can exist independently. **Weak entities** depend on strong entities for their identification and existence.
- Strong entities are identified by a **single rectangle**, while weak entities are represented by a **double rectangle** in an ER diagram.
- Weak entities use a **partial key** together with the primary key of the related strong entity.
- The identifying relationship ensures a proper connection between strong and weak entities.
- Understanding the differences between strong and weak entities improves database design. It helps maintain **data integrity** and reduces redundancy.

THANK YOU